Traditional Methodologies

Software Product and Process Management

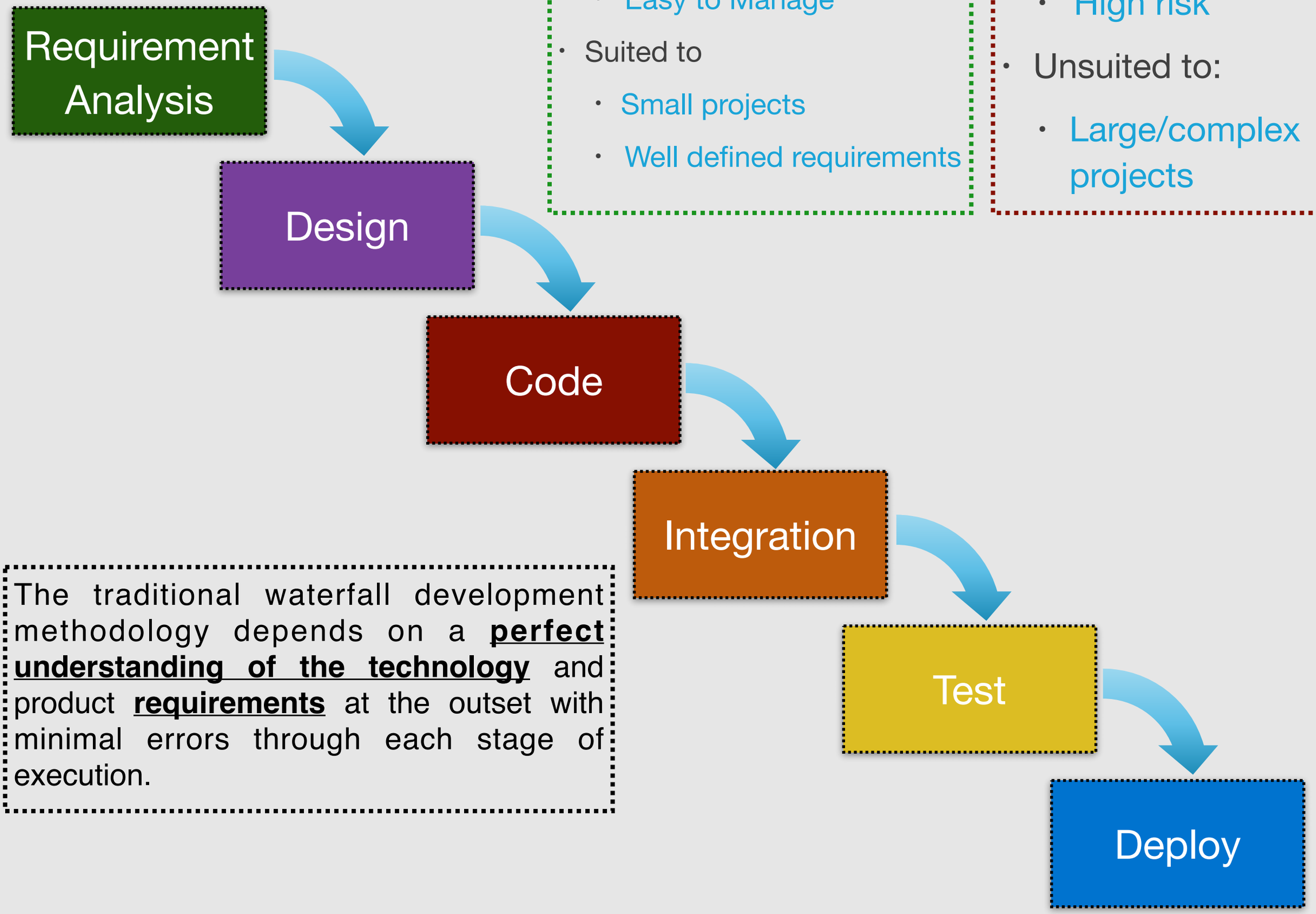
Motivation

- Study on 5,400 large IT projects (>$15 Million)
  - 45% run over budget
  - 7% run over time
  - 56% deliver less value
  - 17% are Black Swans
- Managing projects is hard!
- Opportunity to improve
  - SDLC

Software Development Life Cycles

- SDLCs evolved from other industries
- Manufacturing

Waterfall

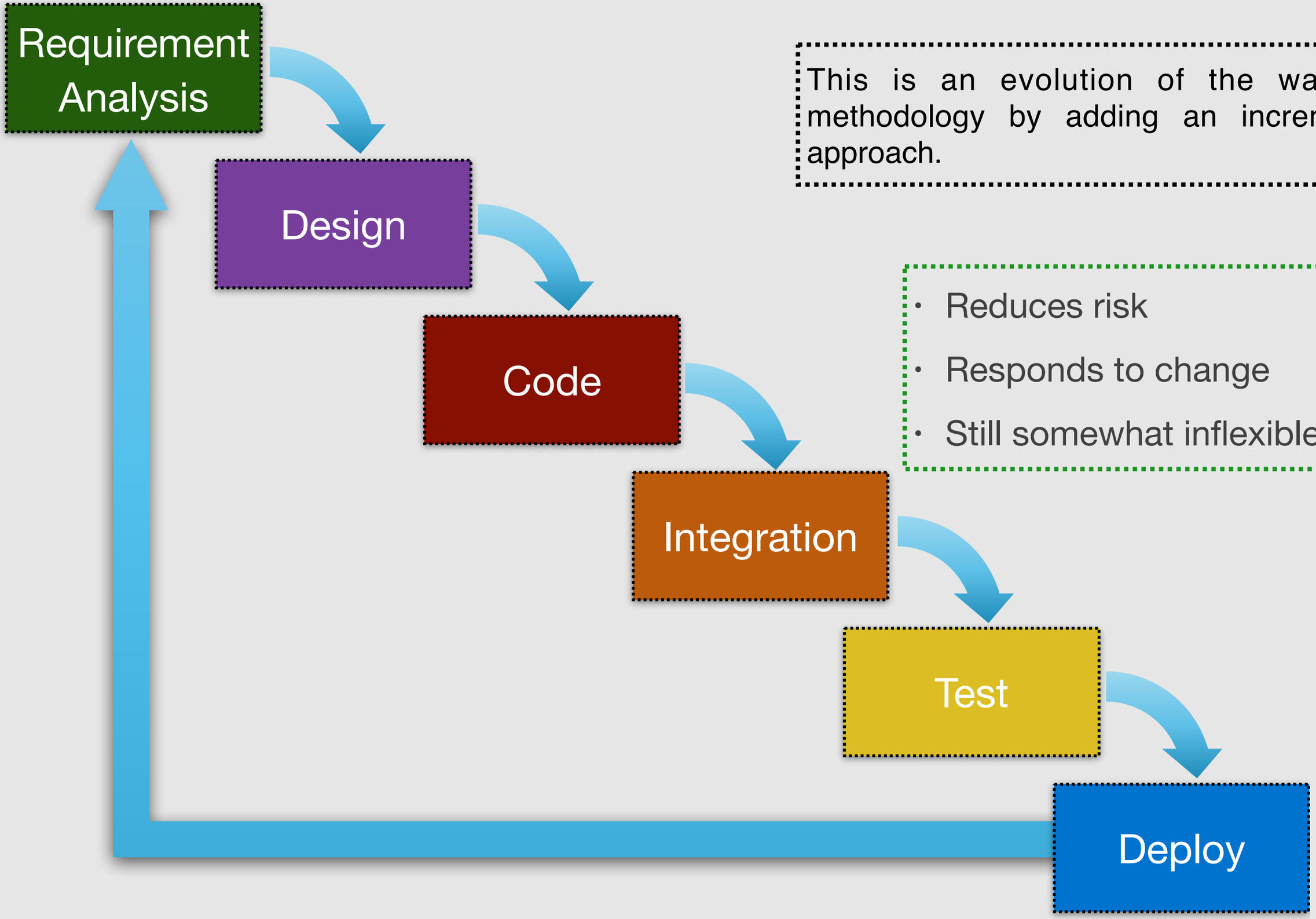


- Rigid model
  - Simple and easy to follow
  - Easy to Manage
- Suited to
  - Small projects
  - Well defined requirements

- Inflexible Model
  - Changes are difficult
  - High risk
- Unsuited to:
  - Large/complex projects

The traditional waterfall development methodology depends on a **perfect understanding of the technology** and product **requirements** at the outset with minimal errors through each stage of execution.

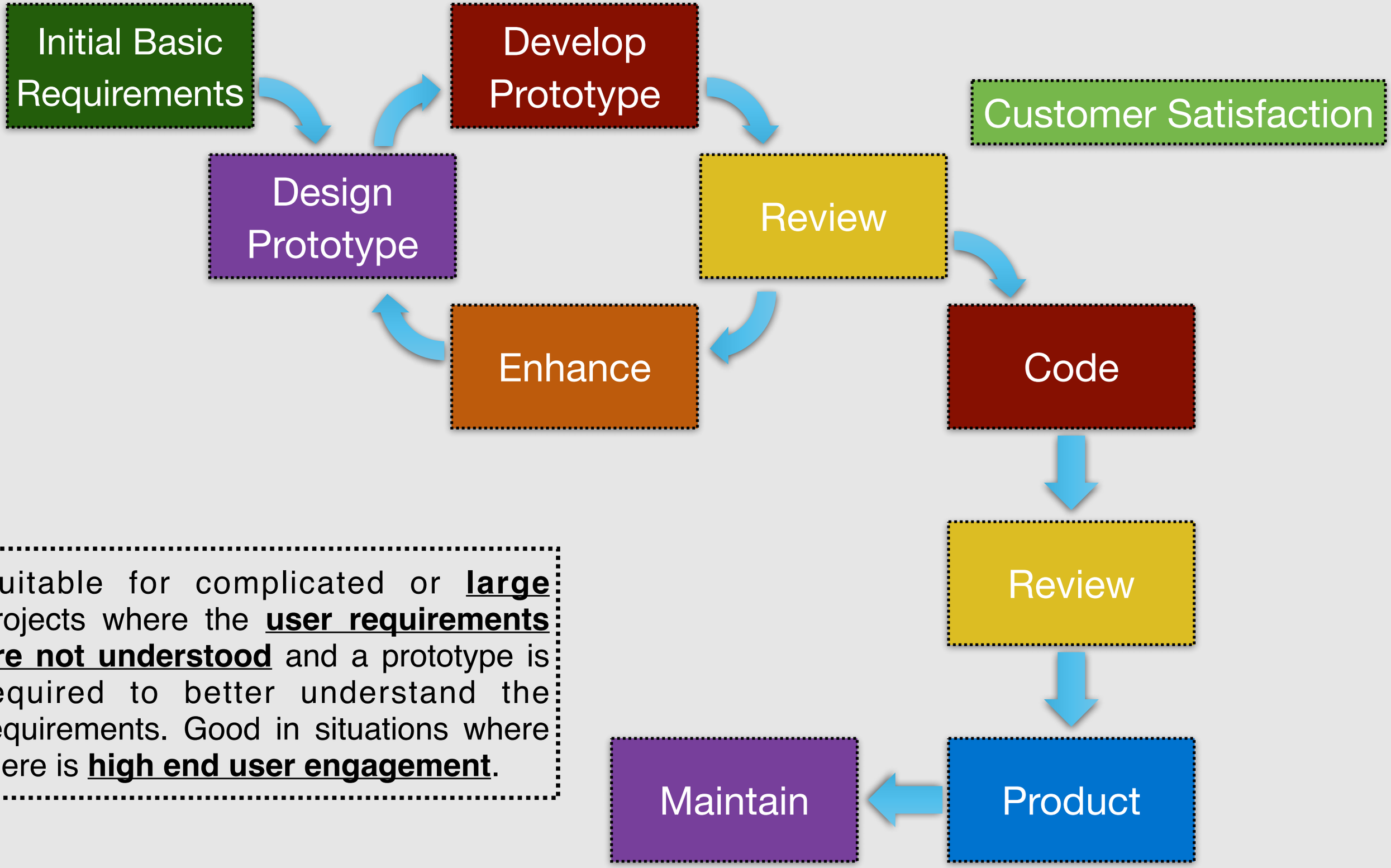
Incremental



This is an evolution of the waterfall methodology by adding an incremental approach.

- Reduces risk
- Responds to change
- Still somewhat inflexible

Prototyping



Suitable for complicated or **large** projects where the **user requirements are not understood** and a prototype is required to better understand the requirements. Good in situations where there is **high end user engagement**.

Prototyping - Advantages

Advantages

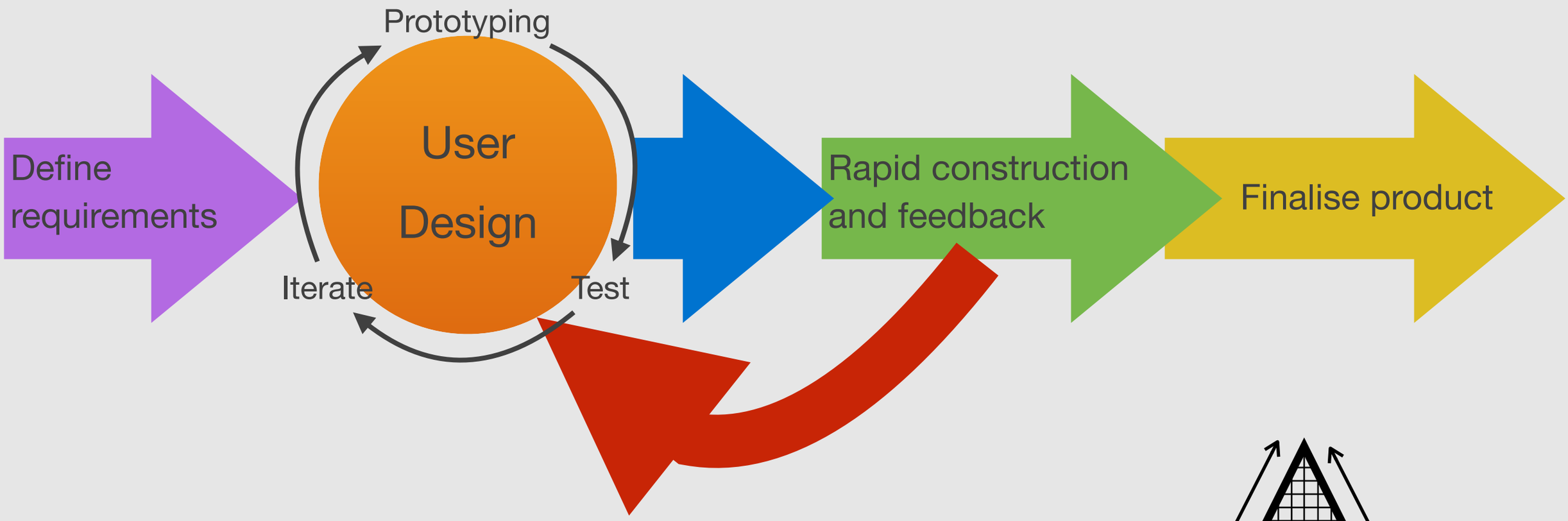
- Increased user involvement in the product even before its implementation.
- Since a working model of the system is displayed, the users get a better understanding of the system being developed.
- Reduces time and cost as the defects can be detected much earlier.
- Quicker user feedback is available leading to better solutions.
- Missing functionality can be identified easily.
- Confusing or difficult functions can be identified.

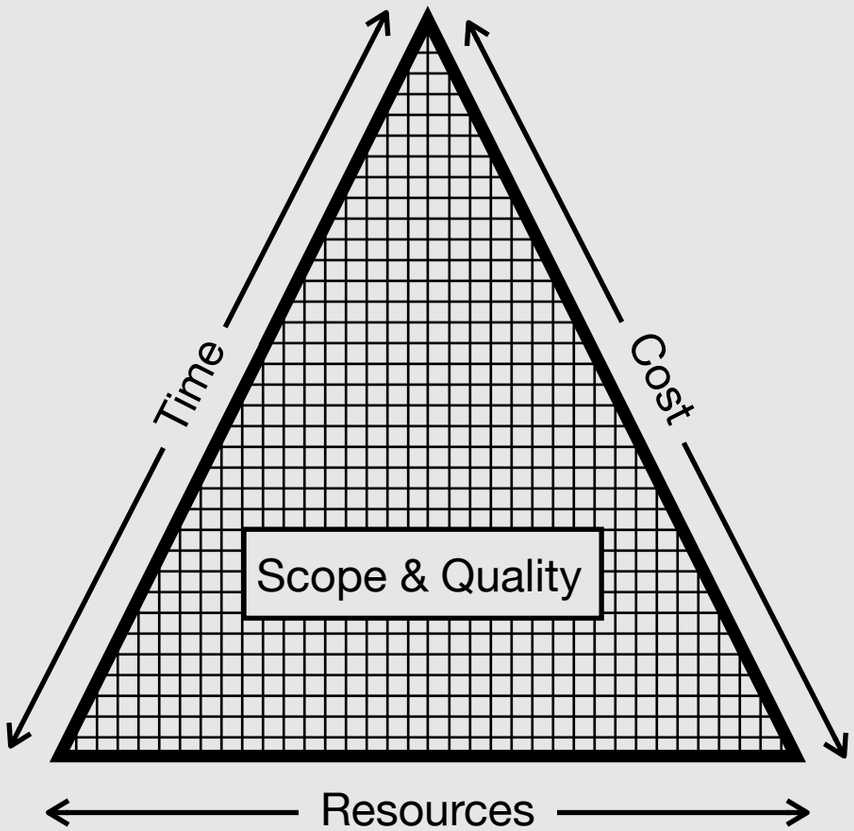
Prototyping - Disadvantages

Disadvantages

- Risk of confusion between between prototypes and actual systems
- Technical difficulties using already development prototypes for the basis of actual systems, may not be technical feasible or may include defects that later will be difficult and cost more to fix.
- Risk of not performing enough requirements analysis because of the fixation with the prototype.
- Prototypes and therefore actual systems can easily have feature creep and move away from original plans

Rapid Application Development





Rapid Application Development - Advantages

Advantages

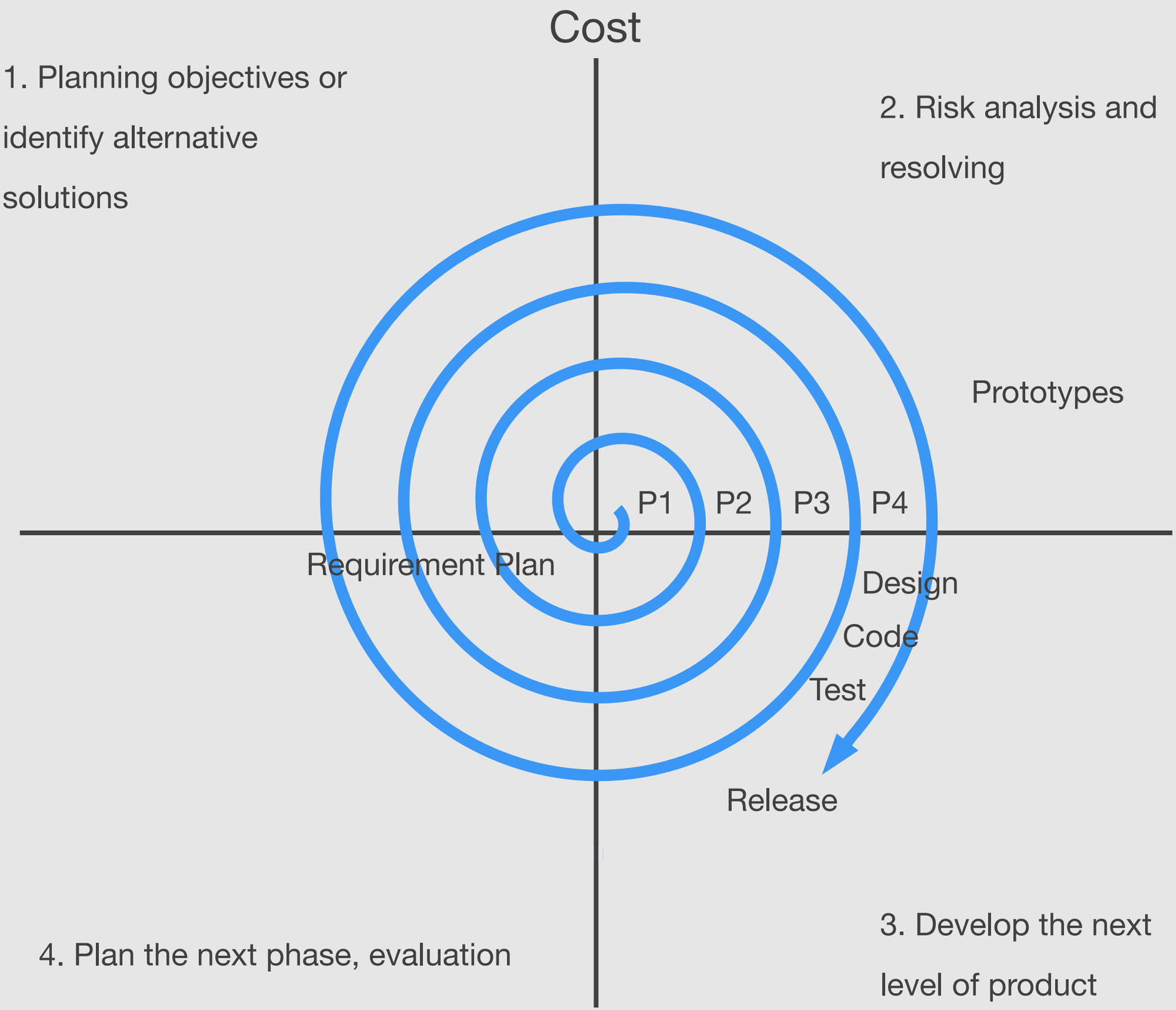
- Features delivered early.
- Speed – it’s fast and concentrates on the minimal deliverables no zombie features.
- Cost effective due to speed, ie. no time wasted on unwanted features, client is happy
- Developers are happy too as they have more contact with the clients to keep them satisfied.

Rapid Application Development - Disadvantages

Disadvantages

- Hard to scale beyond a single team.
- Heavy commitment from the client
- Interface-focus - client judges quality on what they can see and interact with therefore development give the front-end more focus and forego best practice and attention to detail on the backend

Spiral


Cost
1. Planning objectives or identify alternative solutions
2. Risk analysis and resolving
Prototypes
P1
P2
P3
P4
Requirement Plan
Design
Code
Test
Release
4. Plan the next phase, evaluation
3. Develop the next level of product

Spiral - Advantages

Advantages

- **Suitable for large projects:** Spiral models are recommended when the project is large, bulky or complex to develop.
- **Risk Handling:** There are a lot of projects that have un-estimated risks involved with them. For such projects, the spiral model is the best SDLC model to pursue because it can analyze risk as well as handling risks at each phase of development.
- **Customer Satisfaction:** Customers can witness the development of product at every stage and thus, they can let themselves habituated with the system and throw feedbacks accordingly before the final product is made. Space for customer feedback.
- **Requirements flexibility:** All the specific requirements needed at later stages can be included precisely if the development is done using this model.

Spiral - Disadvantages

Disadvantages

- Can be costly to run and not suited to smaller projects.
- Can be difficult to control time and therefore budget, could run indefinitely, i.e not clear definition of done set out.
- Requires risk assessment expertise.
- Moderate documentation required while following each of the phases.

Traditional Methodologies

- Take a look at the project requirements. Are the requirements clear? If project requirements are unclear or tend to change, choose the agile methodology. And, the traditional method fits best in a situation where the conditions are clearly defined and well understood from the first go.

- Consider the technology involved in the project. The traditional project management methodology is more appropriate if no new technology or tools are applied. Agile methods allow more space for experimentation with the latest technology as it is more flexible than the traditional approach.

- Is the project prone to unwanted risks and threats? Considering the rigid nature of the traditional methodology, it's not advisable to go with this methodology. However, risks can be addressed sooner in the agile approach, and it seems like a better option in terms of risk management.

- Another critical factor is the availability of resources. The traditional approach works best with significant and complex teams and projects. Whereas an agile team usually consists of a limited number of experienced team members.

Benefits of Agile

- high quality
- productivity
- better business value
- lower costs
- quicker time-to-market speeds.
- adapts quicker to dynamic business needs

Spotify Engineering Culture

- Two short video on Spotify